

Distinct Element Counting

Estimating F_0 with a Minimum-Value Sketch

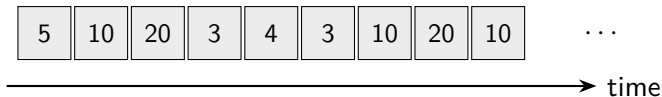
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What is F_0 ?

A stream of items arrives from a universe $[n]$.



$$F_0 = \#\{5, 10, 20, 3, 4\} = 5$$

Goal: estimate F_0 using *sublinear* space.

Why not just store a set?

$\{5, 10, 20, 3, 4, \dots\}$

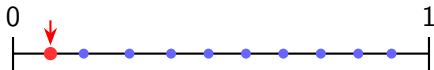
$\Theta(F_0)$ memory

- ▶ Unique visitors to a website: billions per day
- ▶ Distinct IPs in network traffic: millions per second
- ▶ Distinct queries in a search log: terabytes

Question: Can a tiny sketch give a good estimate?

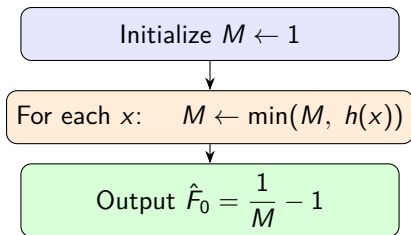
Key Idea

Hash each element to a *random point* in $[0, 1]$.



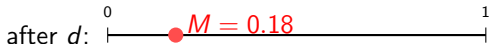
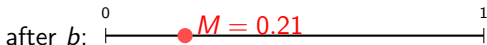
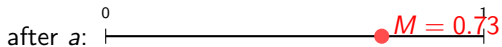
With $d = 10$ uniform points in $[0, 1]$, the minimum $M \approx \frac{1}{d} = 0.1$.

Minimum-Value Sketch



Space: one number M + one hash function.

Example: Watching M Shrink



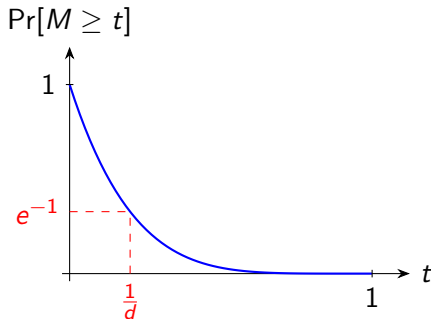
Final: $M = 0.18$, $\hat{F}_0 = 1/0.18 - 1 \approx 4.6$

(true $F_0 = 4$)

Why does it work?

Let $d = F_0$. The minimum of d uniform samples satisfies:

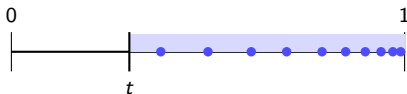
$$\Pr[M \geq t] = (1 - t)^d$$



Step 1: Survival function of M

$M = \min(U_1, \dots, U_d)$ where $U_i \sim \text{Unif}[0, 1]$.

$M \geq t$ means every point lands to the right of t :



$$\Pr[M \geq t] = \Pr[U_1 \geq t] \cdots \Pr[U_d \geq t] = (1 - t)^d$$

$$\mathbb{E}[M] = \int_0^1 \Pr[M \geq t] dt$$

M is the length of the segment $[0, M]$. So:

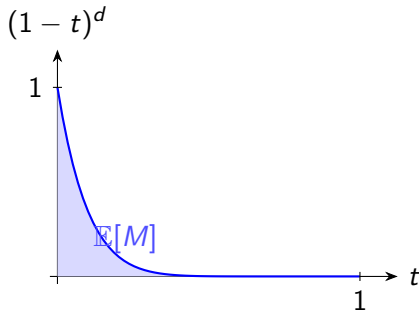
$$M = \int_0^1 \mathbf{1}[t \leq M] dt$$



Take expectations and swap $\int$ and $\mathbb{E}$:

$$\mathbb{E}[M] = \mathbb{E}\left[\int_0^1 \mathbf{1}[t \leq M] dt\right] = \int_0^1 \mathbb{E}[\mathbf{1}[t \leq M]] dt = \int_0^1 \Pr[M \geq t] dt$$

Step 2: Computing $\mathbb{E}[M]$



$$\mathbb{E}[M] = \int_0^1 (1-t)^d dt = \left[-\frac{(1-t)^{d+1}}{d+1} \right]_0^1 = \frac{1}{d+1}$$

Variance of M

Recall $\mu = \mathbb{E}[M] = \frac{1}{d+1}$.

Claim (Problem Set):

$$\text{Var}(M) = \frac{d}{(d+1)^2(d+2)} \leq \frac{1}{(d+1)^2}.$$

Repeat $3/\epsilon^2$ times and average. The variance becomes

$$\frac{\epsilon^2}{3(d+1)^2}$$

Chebyshev: single sketch accuracy

Chebyshev's inequality:

$$\Pr\left[|M - \mu| \geq \epsilon \frac{1}{d+1}\right] \leq \frac{\text{Var}(M)}{\epsilon^2(d+1)^2} \leq 1/3.$$

Median of means trick

Each estimate gives a $2/3$ probability of being within $(1 \pm \epsilon)$ of $1/(d + 1)$.

Take the median of 100 independent estimates. If the median is bad, then at least 50 estimates are bad. This happens with probability at most

$$1/3^{50}.$$

Complexity

Property	Value
Passes over stream	1
Space (single sketch)	$O(\log n)$ bits
Space (ϵ, δ) -approx	$O\left(\frac{\log(1/\delta)}{\epsilon^2} \log n\right)$ bits
Update time	$O(1)$ per item

HyperLogLog (2007) achieves the same guarantee with space $O\left(\frac{1}{\epsilon^2} \log n\right)$ bits — no extra $\log(1/\delta)$ factor.